Ethan Lin Wong

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Education

University of California, Irvine — Expected Dec. 2026
Master of Data Science — *Irvine, CA*

University of California, Irvine — June 2025
Bachelor of Science in Data Science — *Irvine, CA*

Technical Skills

Machine Learning: Deep Neural Networks (CNN, RNN), Reinforcement Learning, Supervised & Unsupervised Learning, Logistic Regression (GLM), Random Forest, XGBoost, Scikit-learn, Model Explainability (SHAP)
Programming: Python, R, SQL (PostgreSQL/PostGIS, MySQL), C++, Bash
Libraries: PyTorch, TensorFlow, Keras, NumPy, Pandas, GeoPandas, SciPy, Matplotlib, Seaborn, Plotly
Data & Statistics: Signal Processing (FFT), Cross-Validation, Time Series Analysis, Bayesian Inference, A/B Testing
Tools: Docker, Git/GitHub, Tableau, Jupyter Notebooks, LaTeX

Experience

Emission Modeling Intern — Sept. 2026 – Present
California Air Resources Board, Air Quality Planning and Science Division — *Riverside, CA*

- Designed system architecture diagrams and code-flow models tracing airport operational inputs (flight volumes, APU usage) to daily fuel consumption (gallons/day) and criteria pollutant emissions (NOx, VOC, CO, PM2.5, PM10, CO2).
- Refactored legacy VBA emission models into modular Python data pipelines, processing non-military, military, and agricultural inventories into standardized datasets supporting CARB's CEPAM projection system.
- Developed interactive web-based reporting tools (HTML/JS) and automated build scripts to dynamically visualize sector-specific emission profiles and support internal data review.

Research Assistant — July 2026 – Present
Fortin Lab, University of California, Irvine — *Irvine, CA*

- Built interpretable models in R (logistic regression on frequency-band features) that predict from rat hippocampal LFP recordings whether an odor sequence was in order, reaching 73% mean balanced accuracy across 5 rats.
- Implemented Meta-Fusion multimodal architectures to fuse single unit spike trains and LFP time series data, analyzing cross-modality neural feature dynamics and classification performance.

Statistics Tutor — Sept. 2022 – Aug. 2023
Pasadena City College — *Pasadena, CA*

- Guided students through model diagnostics, regression assumptions, and parameter estimation techniques to build strong foundational skills in data modeling and quantitative analysis.

Projects

Stock Directional Forecasting — Feb. 2026 – Mar. 2026

- Engineered 20 features (RSI, MACD, rolling volatility) from 10 years of OHLCV data to evaluate signal-to-noise ratios across ARIMA, Logistic Regression, and Random Forest models.
- Built a Soft Voting Ensemble in R achieving 57.3% precision, beating random-walk directional baselines by 14.6%.

Rubik's Cube Reinforcement Learning Solver — Nov. 2025 – Dec. 2025

- Modeled 3x3x3 and 2x2x2 Rubik's Cube environments as Markov Decision Processes (MDPs) across 4.3×10^{19} states, implementing Policy Iteration in Python with dense sticker-matching reward shaping.
- Constructed a non-learning HybridSolver combining greedy hill-climbing, macro-based move applications, and stochastic exploration, achieving up to an 80% solve rate on short scrambles.

Wildfire Smoke Dispersion Modeling | 2025 ICS Project Expo — Jan. 2025 – May 2025

- Architected a spatial-temporal ETL pipeline in Python (GeoPandas, Shapely) and PostgreSQL/PostGIS, querying NOAA satellite data to match fire origins with smoke plumes via spatial buffers and temporal joins.
- Integrated Open-Meteo API weather streams to construct environmental feature arrays, training XGBoost regressors ($R^2 = 0.45$) with SHAP analysis to identify top predictors of smoke spread.